

# Unit 2 – Software Project Scheduling

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# Agenda

- ▣ Objectives of activity planning
- ▣ Work breakdown structure
- ▣ Bar chart
- ▣ Network planning
  - ▣ Critical path method (CPM)
  - ▣ Program evaluation and review technique (PERT)
  - ▣ Precedence diagramming method (PDM)
- ▣ Shortening project duration
- ▣ Identifying critical activities.

Group 1 Presentation

Group 2 Presentation

# Network Planning Model

- These project scheduling techniques model the project's activities and their relationships as a network.
  - In the network, time flows from left to right.
  - There are two best techniques:
    - CPM(Critical Path Method)
    - PERT(Program Evaluation Review Technique)
  - Both use the “**Activity on Arrow**” approach to visualize the project as a network.
    - Activities are drawn as arrows joining circles,or nodes which represent the possible start or completion of an activity or set of activities.
  - More recently a variation on these techniques, called **precedence networks**, has become popular. This method uses **activity-on-node** networks where activities are represented as nodes and the links between nodes represent precedence (or sequencing) requirements.

# Network Planning Model

- Formulating a Network Model
  - The **first stage** in creating a network model is to **represent the activities and their interrelationships as a graph.**
  - In activity-on-node we do this by representing activities as nodes (boxes) in the graph – the lines between nodes represent dependencies.
- Rules for constructing networks:
  - *A project network should have only one start node.*
  - *A project network should have only one end node.*
  - *A node has duration.*
  - *Links normally have no duration.*
  - *Precedents are the immediate preceding activities.*
  - *Time moves from left to right.*
  - *A network may not contain loops.*
  - *A network should not contain dangles.*

# Network Planning Model

- Formulating a Network Model
  - Labelling conventions:

|                                      |          |                 |
|--------------------------------------|----------|-----------------|
| Earliest start                       | Duration | Earliest finish |
| Activity label, activity description |          |                 |
| Latest start                         | Float    | Latest finish   |

# Network Planning Model

- Adding the time dimension
  - Having created the logical network model indicating what needs to be done and the interrelationships between those activities, we are now ready to start thinking about when each activity should be undertaken.
  - The critical path approach is concerned with two primary objectives:
    - planning the project in such a way that it is completed as quickly as possible; and
    - identifying those activities where a delay in their execution is likely to affect the overall end date of the project or later activities' start dates.
  - The method requires that for each activity we have an estimate of its duration.
    - The network is then analysed by carrying out a **forward pass**, to calculate the earliest dates at which activities may commence and the project be completed, and a **backward pass**, to calculate the latest start dates for activities and the **critical path**.



# Network Planning Model

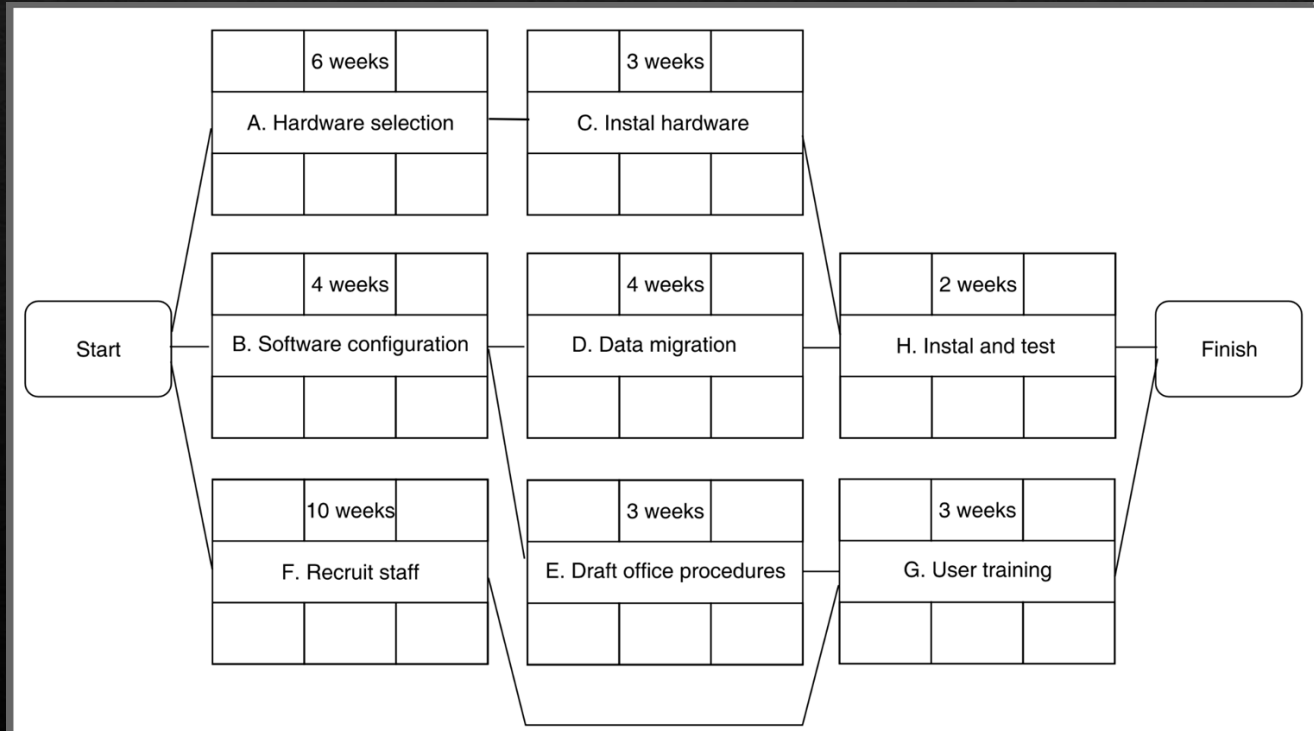
- The forward pass is carried out to calculate the earliest dates on which each activity may be started and completed.

An example project specification with estimated activity durations and precedence requirements

| Activity |                         | Duration (weeks) | Precedents |
|----------|-------------------------|------------------|------------|
| A        | Hardware selection      | 6                |            |
| B        | System configuration    | 4                |            |
| C        | Instal hardware         | 3                | A          |
| D        | Data migration          | 4                | B          |
| E        | Draft office procedures | 3                | B          |
| F        | Recruit staff           | 10               |            |
| G        | User training           | 3                | E, F       |
| H        | Instal and test system  | 2                | C, D       |

# Network Planning Model

□ The precedence network would look like:



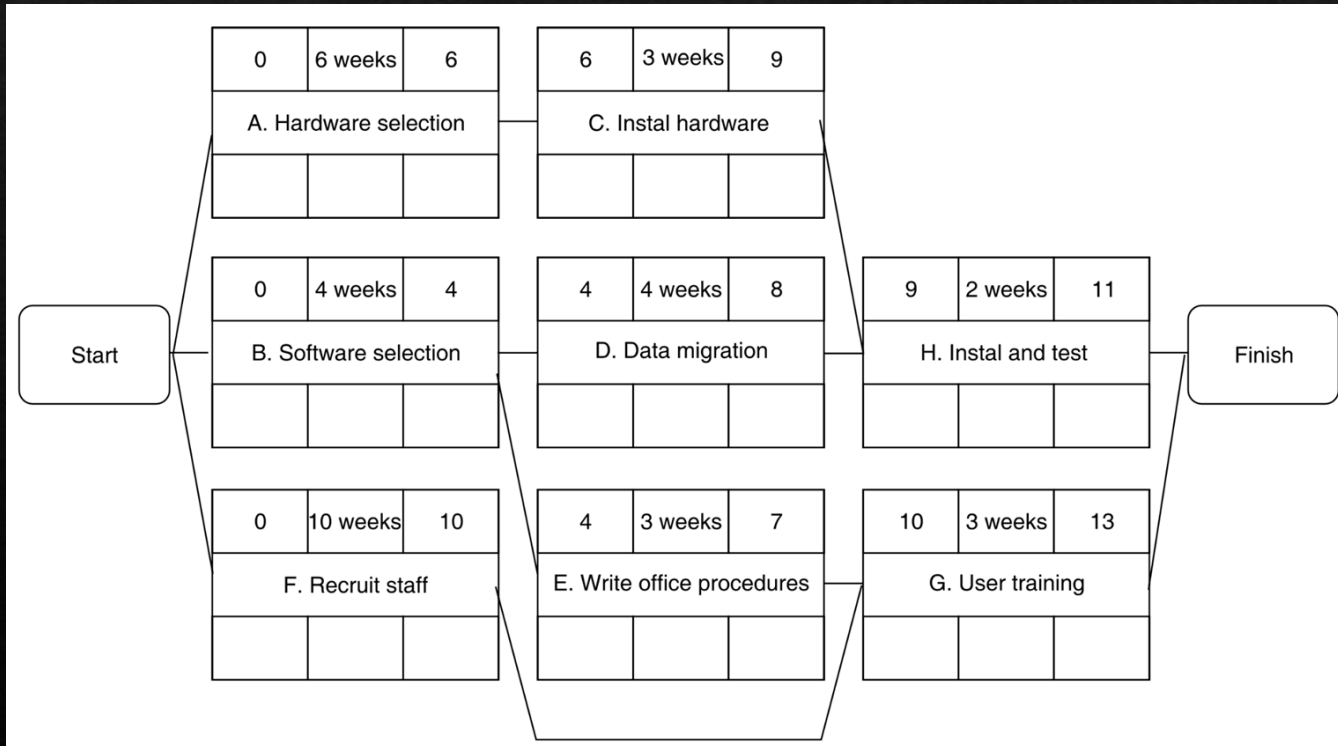


# Network Planning Model (CPM - The Forward Pass)

- During the forward pass, earliest dates are recorded as they are calculated.
- **The forward pass rule:**
  - The earliest start date for an activity is the earliest finish date for the preceding activity.
  - Where there is more than one immediately preceding activity we take the *latest* of the *earliest finish dates* for those activities.

# Network Planning Model (CPM - The Forward Pass)

□ The results of forward pass are shown below:

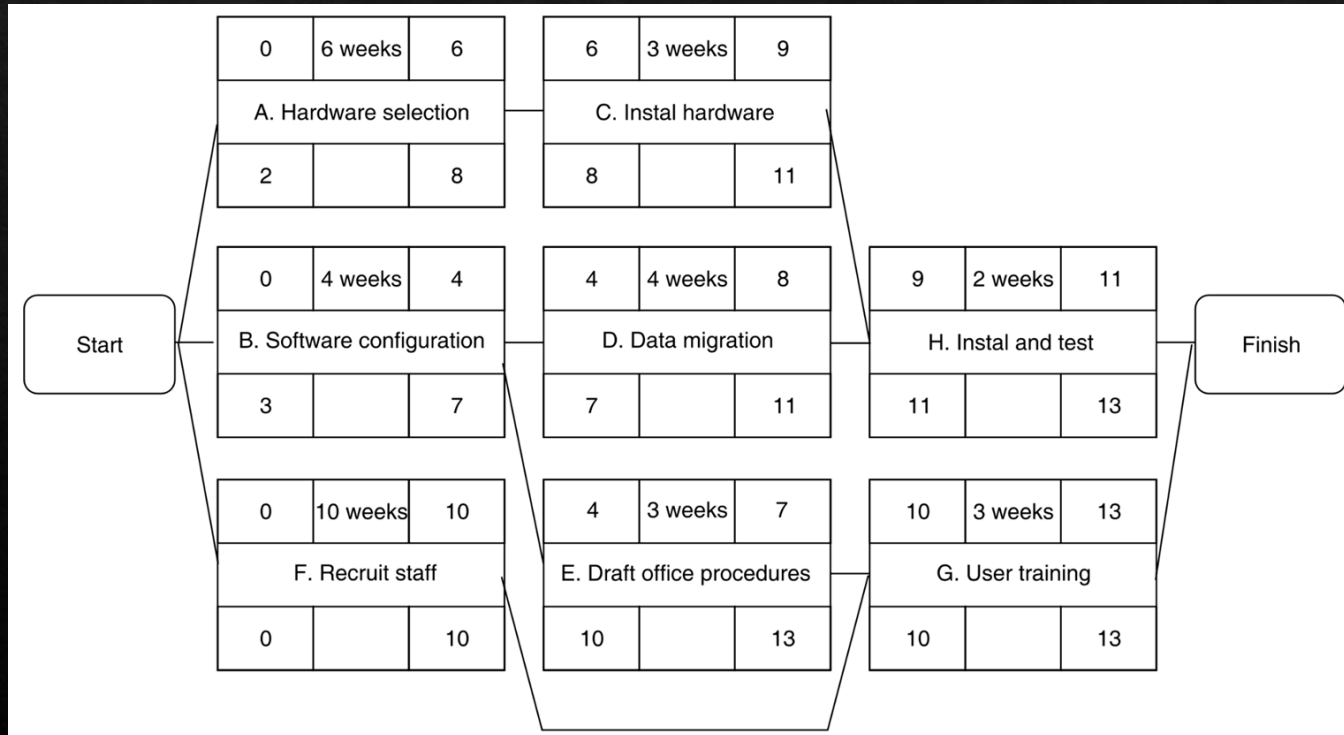


# Network Planning Model (CPM - The Backward Pass)

- The second stage in the analysis of a critical path network is to carry out a *backward pass to calculate the latest date at which each activity may be started and finished without delaying the end date of the project.*
- In calculating the latest dates, we assume that the latest finish date for the project is the same as the earliest finish date – that is, we wish to complete the project as early as possible.
- **The backward pass rule:** The latest finish date for an activity is the latest start date for the activity that commences immediately that activity is complete. Where more than one activity can commence we take the earliest of the latest start dates for those activities.

# Network Planning Model (CPM - The Backward Pass)

□ The results of backward pass are shown below:



# Network Planning Model (CPM)

## □ Critical Path

- *The critical path is the longest path through the network.*
- The difference between an activity's earliest start date and its latest start date is known as the *activity's float*.
  - **Float:** A measure of how much the start or completion of an activity may be delayed without affecting the end date of the project. *Any activity with a float of zero is critical in the sense that any delay in carrying out the activity will delay the completion date of the project as a whole.*
  - **Total Float:** The amount of time that a schedule activity can be delayed or extended from its early start date without delaying the project finish date or violating a schedule constraint.
    - **Total Float or Float =  $LS - ES$  or  $LF - EF$**
  - **Free Float:** The amount of time that a schedule activity can be delayed without delaying the early start date of any successor or violating a schedule constraint
    - **Free Float =  $ES \text{ (of successor)} - EF \text{ (of current)}$**

# Network Planning Model (CPM)

## □ Example 2: Find the critical path

| Task ID | Task                              | Dependency | Estimate (Days) |
|---------|-----------------------------------|------------|-----------------|
| 1       | Collect contacts for publications | -          | 0.5             |
| 2       | Create invite                     | -          | 0.5             |
| 3       | Send out invites                  | 1, 2       | 0.5             |
| 4       | Confirm speakers and panelists    | -          | 1               |
| 5       | Check speaker slides              | 4          | 0.5             |
| 6       | Press release outline             | -          | 1               |
| 7       | Collect assets                    | -          | 1               |
| 8       | Draft press release               | 6          | 1               |
| 9       | Finalize press release            | 8          | 3               |
| 10      | Send out press kit                | 9, 1, 7    | 1               |
| 11      | Implement press conference        | 10, 5, 3   | 0.5             |



# Network Planning Model (PERT)

- PERT: Program Evaluation and Review Technique
- Take account of the uncertainty surrounding estimates of task durations.
- The method is very similar to the CPM technique but, instead of using a single estimate for the duration of each task, PERT requires three estimates:
  - **Most likely time**: the time we would expect the task to take under normal circumstances. We shall identify this by the letter *m*.
  - **Optimistic time**: the shortest time in which we could expect to complete the activity. We shall use the letter *a* for this.
  - **Pessimistic time**: the worst possible time, allowing for all reasonable eventualities but excluding 'acts of God and warfare' (as they say in most insurance exclusion clauses). We shall call this *b*.
- PERT then combines these three estimates to form a single expected duration,  $t_e$ , using the formula

$$t_e = (a + 4m + b) / 6$$



# Network Planning Model (PERT)

- The expected durations are used to carry out a forward pass through a network, using the same method as the CPM technique. *In this case, however, the calculated event dates are not the earliest possible dates but the dates by which we expect to achieve those events.*

# Network Planning Model (PERT)

## □ Example

| Activity | Optimistic ( $a$ ) | Activity durations<br>(weeks). Most likely ( $m$ ) | Pessimistic ( $b$ ) |
|----------|--------------------|----------------------------------------------------|---------------------|
| A        | 5                  | 6                                                  | 8                   |
| B        | 3                  | 4                                                  | 5                   |
| C        | 2                  | 3                                                  | 3                   |
| D        | 3.5                | 4                                                  | 5                   |
| E        | 1                  | 3                                                  | 4                   |
| F        | 8                  | 10                                                 | 15                  |
| G        | 2                  | 3                                                  | 4                   |
| H        | 2                  | 2                                                  | 2.5                 |

# Network Planning Model (PERT)

| Activity | Activity durations (weeks) |                     |                     |                    |
|----------|----------------------------|---------------------|---------------------|--------------------|
|          | Optimistic ( $a$ )         | Most likely ( $m$ ) | Pessimistic ( $b$ ) | Expected ( $t_e$ ) |
| A        | 5                          | 6                   | 8                   | 6.17               |
| B        | 3                          | 4                   | 5                   | 4.00               |
| C        | 2                          | 3                   | 3                   | 2.83               |
| D        | 3.5                        | 4                   | 5                   | 4.08               |
| E        | 1                          | 3                   | 4                   | 2.83               |
| F        | 8                          | 10                  | 15                  | 10.50              |
| G        | 2                          | 3                   | 4                   | 3.00               |
| H        | 2                          | 2                   | 2.5                 | 2.08               |

# Network Planning Model (PERT)

| Activity | Description                             | Predecessors | Optimistic Duration<br>(To) | Pessimistic Duration<br>(Tp) | Most likely Duration<br>(Tm) |
|----------|-----------------------------------------|--------------|-----------------------------|------------------------------|------------------------------|
| 0        | Start Milestone                         | -            | 0                           | 0                            | 0                            |
| A        | Select Technical Staff                  | 0            | 12                          | 18                           | 15                           |
| B        | Site Survey                             | 0            | 6                           | 12                           | 9                            |
| C        | Select Equipments                       | A            | 9                           | 15                           | 12                           |
| D        | Prepare Designs                         | B            | 6                           | 18                           | 9                            |
| E        | Bring Utilities to the Site.            | B            | 18                          | 36                           | 30                           |
| F        | Interview Applicants and Fill Positions | A            | 9                           | 15                           | 12                           |
| G        | Purchase the Equipment.                 | C            | 36                          | 42                           | 36                           |
| H        | Construct the Power Plant               | D            | 42                          | 54                           | 48                           |
| I        | Develop an Information System.          | A            | 6                           | 18                           | 12                           |
| J        | Install the Equipment.                  | H,G,E        | 3                           | 9                            | 6                            |
| K        | Train the Staff to Run the System       | F,J,I        | 3                           | 9                            | 6                            |

# Network Planning Model (PERT)

- Determine critical path
- Compute variance and standard deviation
  - *Need to compute the variance (and std. dev) for activities in Critical Path only.*
- Calculate probability of completing the project in 82 days.